Romane Castel

Software Engineer | LLM agents, model evaluation, and generative AI systems

Zurich, CH | romane.castel.pro@gmail.com | in/romane-castel | github.com/RomaneCastel

Education

2019 – Jun 2020
Switzerland

ETH Zurich | Exchange semester in Machine Learning
Advanced machine learning, Probabilistic AI, Neural networks theory, Reliable and interpretable AI, Data mining, Advanced topics in machine learning. **GPA 5.0/6.0.**

2017 – Jun 2020
France

ENSSAT | MEng in Computer Science
Mathematics, Algorithms, Operating systems, Data structures, Object-oriented programming. **Grade A (GPA 4.0/4.0).**

2014 – Jun 2017
France

Lycée Kerichen | Preparatory classes (CPGE)
Intensive mathematics and physics programme. **Grade A (GPA 4.0/4.0).**

Professional Experience

2026
Zurich, CH

Luware | Software Engineer

- Built and shipped a multi-agent AI assistant for a contact-centre platform, with the agent core in **Python** (**LangGraph**/LangChain), using LLMs served through **Azure AI Foundry**.
- Built the evaluation harness used to test and compare models: it replays assistant tasks in several languages, scores answers with LLM judges, and catches quality regressions early.
- Added confidence scoring and explicit abstention to model outputs, with deterministic validation in code, so that only high-confidence suggestions are applied automatically.
- Built **MCP** tools that let LLM agents act on the platform, with input validation, gateway-based authorization and error handling that never leaks internal details.
- Built an event-driven pipeline that automatically tags customer conversations once transcription completes, with idempotent, retry-safe processing that never overrides manual edits.
- Made model upgrades low-risk with per-customer rollout and automatic fallback, and delivered features end to end across Angular, .NET, Python/FastAPI and Terraform CI/CD.

2025
Zurich, CH

Disney Research Studios | Research Engineer

- Worked closely with researchers to turn their text-to-image and text-to-video models into an internal multimodal generative-AI platform (**PyTorch**, Hugging Face, Ollama), updated with each new model.
- Evaluated generative video outputs across model iterations on a fixed prompt set, combining side-by-side review with shot-to-shot similarity metrics to detect visual incoherence and drive model improvements.
- Curated and cleaned training data; turned research code into a robust tool (React, Python/Flask).

2022
Zurich, CH

Microsoft Switzerland | Software Engineer
Dynamics 365 — Customer Insights

- Integrated and shipped the Copilot Chat experience for an enterprise product (**TypeScript/React**, **Power Virtual Agents**, **Azure**).
- Part of the on-call rotation for an enterprise-scale live service; improved production reliability by investigating and fixing critical infrastructure and application weaknesses.

2020
Copenhagen, DK

Microsoft Denmark | Software Engineer
Dynamics 365 — Customer Insights

- Built large-scale distributed data pipelines in **C#** on **Azure** (**Apache Spark**, **Hadoop**) to unify customer profiles from complex datasets.
- Replaced a legacy polling system with event-driven push notifications via **SignalR**, cutting status-call traffic by 80%.

2020
Zurich, CH

DeepCode | Research Intern, Deep Learning (MSc thesis)

- Trained and evaluated sequence-to-sequence and **GPT-2** models (**Python**, **PyTorch**) to repair **C++** null-pointer bugs, framed as graph transformations on code graphs: predicting which nodes and edges to add, remove, or move.
- Built the core AI component of an automated bug-fixing tool; thesis awarded Grade A.

2019
London, UK

Amazon | Software Engineer Intern
Amazon Transportation Services

- Deployed an ML model to production for cost prediction and outlier detection (**scikit-learn** on **AWS SageMaker**, **Django** backend).
- Developed and mathematically optimised a cost allocation algorithm for Amazon finances.

Projects

2020
ETH Zurich

Neural Net Robustness Verifier

- Built an automatic verifier that certifies whether a neural network is robust to L-infinity perturbations on input images, returning an answer in under two minutes on both fully connected and convolutional networks (**Python**, **PyTorch**). Maximum grade, 6/6.

Skills

Programming	Python, C#/.NET, C/C++, TypeScript/JavaScript
ML & Evaluation	LLM agents (LangGraph/LangChain, MCP), LLM-as-judge evaluation, PyTorch, Hugging Face, Azure AI Foundry, regression test suites, scikit-learn
Infra & MLOps	CI/CD (Azure DevOps), Terraform/Terragrunt, Azure, AWS (SageMaker), OpenTelemetry, feature-flagged rollouts
Data & Systems	Apache Spark, Hadoop, RavenDB, event-driven architecture, SignalR, FastAPI, Flask, REST APIs
Front end	React, Angular
Languages	French (native), English (fluent)
Misc	Piano, running, climbing, skiing, hiking, crochet